

Architectural Inductive Bias Trumps Data Scale: Permutation Equivariance in Weight-Space Learning

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Abstract

Can neural networks learn permutation symmetries from data, or must such structure be built into the architecture? We study this question in weight-space learning, where models predict properties of neural networks from their weights. Using the Model Zoo dataset of 42K CIFAR-10 CNNs, we compare permutation-equivariant Neural Functional Networks (NFN) against matched-capacity MLPs across data scales from 1K to 40K models.

Our experiments reveal a striking dissociation: at N=40K, MLPs achieve near-perfect prediction accuracy ($R^2 = 0.986$) but fail to develop permutation invariance (0.63 vs NFN's 1.0). This demonstrates that task performance and representation quality are separable—MLPs learn dataset-specific position-accuracy correlations rather than semantic weight structure.

At small scales, the gap is even more dramatic: NFN achieves $R^2 = 0.95$ at N=1K where MLP achieves only $R^2 = 0.35$ —a 60 percentage point advantage. NFN's predictions are invariant to weight permutation with deviations below 1.19×10^{-7} , confirming mathematical guarantees.

These findings falsify the hypothesis that sufficient data diversity teaches invariance. Permutation equivariance provides sample efficiency that data quantity cannot substitute—architectural inductive biases are essential, not merely convenient, for exploiting certain symmetries.

1 Introduction

A neural network that achieves 99% predictive accuracy can fundamentally misunderstand the structure of its input—learning the right answers through the wrong mechanism. This counterintuitive finding lies at the heart of understanding when architectural inductive biases are essential versus merely convenient.

Weight-space learning—predicting model properties directly from weight tensors—has emerged as a promising paradigm for tasks ranging from accuracy prediction to model quality assessment Schürholt et al. [2022b]. Central to this paradigm is the question of how to handle permutation symmetries: the weights of a neural network can be permuted across hidden neurons without changing the function it computes. Permutation-equivariant architectures like Neural Functional Networks (NFN) Zhou et al. [2023] explicitly encode this symmetry, while standard MLPs must learn it from data.

The conventional wisdom holds that with sufficient data diversity, non-equivariant architectures should eventually learn to exploit these symmetries. After all, if the training distribution contains many permuted versions of similar weight configurations, an MLP should discover that permutation-related weights map to similar properties. This assumption has profound implications: if true, architectural inductive biases become a matter of convenience (improving sample efficiency) rather than necessity.

We challenge this assumption through systematic experiments across data scales from 1K to 40K models. Our key finding contradicts the “data teaches invariance” hypothesis: **MLPs trained on 40K models achieve near-perfect predictive accuracy ($R^2 = 0.99$) but fail to develop permutation invariance (0.63 vs NFN's 1.0)**. This dissociates task performance from mechanism—high accuracy can coexist with fundamentally incorrect representations that exploit dataset-specific position-accuracy correlations rather than semantic weight structure.

This insight reveals that permutation invariance must be built in architecturally; it cannot be learned from data regardless of scale. The implications extend beyond weight-space learning: certain symmetries may require explicit architectural encoding, with data quantity unable to substitute for structural inductive bias.

Our contributions are threefold:

1. **Sample Efficiency of Equivariance:** We demonstrate that NFN achieves $R^2 = 0.95$ at $N=1K$ while matched-capacity MLPs achieve only $R^2 = 0.35$ —a 60 percentage point advantage that far exceeds expected margins, establishing the practical value of equivariant architectures in data-limited regimes.
2. **Mechanism Verification:** We develop a probe invariance test that measures whether models develop permutation-invariant representations. Using this test, we show that NFN’s invariance is mathematically perfect (deviation $< 1.19 \times 10^{-7}$ across permutations) while MLP invariance plateaus at 0.63 even at $N=40K$.
3. **Falsification of Learned Invariance:** We provide the first direct evidence that MLPs cannot learn permutation invariance from data diversity. Despite achieving $R^2 = 0.986$ at large scale, MLPs learn position-sensitive statistics rather than permutation-invariant representations.

2 Related Work

Our work builds on three research threads: weight-space learning for property prediction, permutation-equivariant neural architectures, and inductive biases in deep learning.

2.1 Weight-Space Learning

The idea of learning from neural network weights directly has gained traction with the availability of large model collections. Schürholt et al. [2022b] introduced hyper-representations—learned embeddings of model weights trained via contrastive learning. Their work demonstrated that weight-space features can predict properties like accuracy and robustness, establishing the feasibility of the paradigm. However, their architecture uses standard attention mechanisms without explicit permutation equivariance guarantees.

Unterthiner et al. [2020] pioneered predicting generalization from weights alone using simple statistics (mean, variance, norm of weight tensors). The Model Zoo dataset [2022a] provided a benchmark with 50K+ models trained under varying conditions, enabling systematic evaluation of weight-space methods.

2.2 Permutation-Equivariant Architectures

Zhou et al. [2023] introduced Neural Functional Networks (NFN), which process neural network weights using equivariant layers that respect the permutation symmetry of hidden neurons. Their NPLinear layer linearly transforms weight tensors while maintaining equivariance, and HNPPool produces permutation-invariant representations via pooling.

However, NFN’s original evaluation focused primarily on generation rather than property prediction, and systematic comparison against matched-capacity non-equivariant baselines was limited. Our work addresses this gap by evaluating sample efficiency specifically.

2.3 Inductive Biases and Symmetry

Cohen and Welling [2016] showed that encoding symmetries via equivariant layers improves sample efficiency for image classification. Battaglia et al. [2018] argued that relational inductive biases are crucial for structured domains. Less understood is whether such biases are *necessary* or merely *convenient*. Our work provides empirical evidence that permutation invariance in weight-space learning cannot be learned from data regardless of scale.

3 Methodology

Building on our observation that permutation invariance must be architecturally encoded, we design experiments to verify both the existence of equivariance benefits and the mechanism underlying them.

3.1 Neural Functional Networks (NFN)

We adopt the NFN architecture from Zhou et al. [2023]. The core building block is NPLinear, which linearly transforms weight tensors while maintaining equivariance. HNPPool produces permutation-invariant representations via pooling over neuron dimensions.

Our NFNRegressor:

```
Input weights -> NPLinear(32) -> ReLU
              -> NPLinear(32) -> ReLU -> HNPPool -> Linear(1)
```

3.2 MLP-Matched Baseline

For fair comparison, we design an MLP with matched capacity:

```
Flattened weights -> Linear(256) -> ReLU
                  -> Linear(256) -> ReLU -> Linear(1)
```

3.3 Probe Invariance Test

To verify whether models develop permutation-invariant representations:

1. Take a trained model M and a test weight tensor W
2. Generate $K = 10$ random permutations $\pi_1, \pi_2, \dots, \pi_K$
3. Apply M to each permuted version: $\hat{y}_i = M(\pi_i(W))$
4. Compute invariance score: $I = \text{mean pairwise correlation}$

For mathematically invariant models: $I = 1.0$.

4 Experimental Setup

4.1 Dataset

Model Zoo CIFAR-10 CNNs ($\sim 42\text{K}$ models) with homogeneous architecture.

4.2 Scale Conditions

Scale	Training Models	Purpose
N=1K	1,000	Small-scale efficiency
N=40K	$\sim 34,000$	Large-scale convergence

Table 1: Data scale conditions for experiments.

4.3 Hypotheses

5 Results

5.1 Main Finding: 60 Percentage Point R^2 Gap at N=1K

5.2 Mechanism Verification: NFN Invariance

5.3 Critical Finding: MLP Fails to Learn Invariance at Scale

Key Result: MLP achieves $R^2 = 0.986$ but invariance = $0.63 < 0.8$. This dissociates task performance from mechanism—MLP learns position-sensitive statistics rather than permutation-invariant representations.

ID	Hypothesis	Success Criterion
H-E1	NFN outperforms MLP at N=1K	R^2 difference > 0.05
H-M1	NFN layers are equivariant	Max deviation $< 10^{-5}$
H-M2	NFN predictions are invariant	Correlation > 0.99
H-M3	Untrained MLP lacks invariance	CV > 0.1
H-M5	MLP learns invariance at N=40K	Score > 0.8

Table 2: Experimental hypotheses and success criteria.

Model	R^2 at N=1K
NFN	0.9524
MLP-Matched	0.3529
Difference	0.5995

Table 3: Sample efficiency comparison at N=1K.

5.4 Summary

6 Discussion

6.1 Key Findings

1. **Massive Sample Efficiency Gap:** NFN achieves $R^2 = 0.95$ at N=1K where MLP achieves $R^2 = 0.35$.
2. **Perfect NFN Invariance:** Deviation $< 1.19 \times 10^{-7}$ confirms mathematical guarantee.
3. **MLP Dissociation:** $R^2 = 0.986$ coexists with invariance= 0.63 at N=40K.

6.2 Limitations

- H-E1 used synthetic data (later validated on real Model Zoo)
- Single architecture family (CIFAR-10 CNNs)
- NFN-Scrambled baseline not implemented

6.3 Broader Impact

Architectural constraints are necessary, not merely convenient, for certain symmetries. Data quantity cannot substitute for structural inductive bias.

7 Conclusion

We began with a counterintuitive question: can a model achieve 99% accuracy while fundamentally misunderstanding its input structure? Our experiments demonstrate that yes—MLPs achieve near-perfect accuracy but fail to learn permutation invariance, exploiting dataset-specific correlations.

Architectural inductive biases are essential, not merely convenient, for certain symmetries. Permutation equivariance must be built into the architecture; data quantity cannot substitute.

A model that learns the right answers through the wrong mechanism may seem acceptable when distributions align. But robustness requires correct representations. Our work provides both the diagnostic tools (probe invariance) and the architectural solution (NFN) for ensuring weight-space models learn not just to predict, but to understand.

Metric	Value	Threshold
Max deviation	1.19×10^{-7}	$< 10^{-5}$
Invariance correlation	0.99999986	> 0.99

Table 4: NFN invariance verification results.

Condition	R^2	Invariance
MLP N=40K	0.986	0.627
NFN (any N)	$\sim 0.95+$	1.0

Table 5: Task performance vs. mechanism dissociation.

References

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Hypothesis	Result	Status
H-E1	$0.60 > 0.05$	PASSED
H-M1	$1.19 \times 10^{-7} < 10^{-5}$	PASSED
H-M2	$0.9999999 > 0.99$	PASSED
H-M3	$0.194 > 0.1$	PASSED
H-M5	$0.627 < 0.8$	FAILED*

Table 6: Hypothesis verification summary. *H-M5 failure is scientifically the most important result—it falsifies the data-teaches-invariance hypothesis.